

# AVERAGE

- ▶ Average of Different types of numbers
- ▶ Consecutive Numbers
- ▶ replacement
- ▶ Addition
- ▶ Weighted average


$$\text{Average Formula} = \frac{\text{Total Sum of All Numbers}}{\text{Number of Item in the Set}}$$




# Different Types of Numbers

- Natural-
  - Odd -
  - Even-
  - Squares-
  - Cubes-
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- 1. What is the average of the first 80 natural numbers?**
  - 2. What is the average of the first 75 natural numbers?**
  - 3. What is the average of the first 25 even numbers?**
  - 4. What is the average of the first 40 odd numbers ?**
  - 5. Find the average of squares of the first 20 natural numbers**
  - 6. Find the average of cubes of the first 10 natural numbers**



# Consecutive Numbers

**Q. If the average of the sum of the 4 consecutive Even numbers is 27. Find the largest of these numbers?**

**a.20**

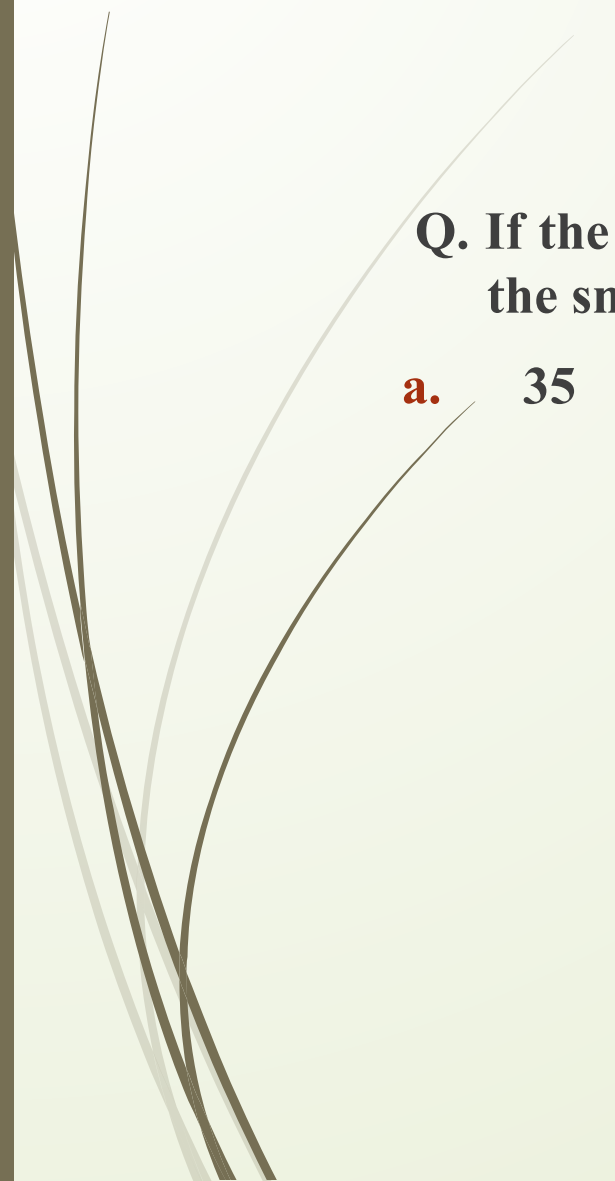
**b.24**

**c.32**

**d 30**



# Consecutive Numbers



**Q. If the average of the sum of the 4 consecutive Odd numbers is 32. Find the smallest of these numbers?**

- a.** 35      **b.** 29      **c.** 33      **d.** 31



# Practice

- **Average of 6 consecutive odd number is 40. Find the largest number**
  - **The average of 8 consecutive even number is 27. Find the largest of these numbers**
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# Replacement



**1. The average weight of 8 men increased by 2 kg when one of them whose weight is 45 kg replaced by a new person. Find the weight of the new person.**



# Replacement

**2. The average age of 10 men increased by 1.5 years when two of them whose ages are 25 and 20 years replaced by two new men. Find the average of those two new men**







# Replacement

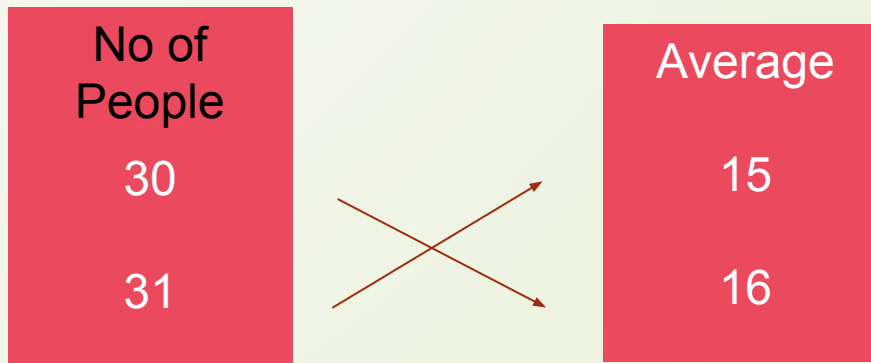
**3. The average age of 8 men increased by 2 years when two of them whose ages are 21 and 23 years are replaced by two new men. The average age of two new men-**



# Addition

1. The average age of 30 students of a class is 15 years. If the age of the teacher is also included, the average increased by 1 year. Find the age of the teacher

- a. 47      b. 43      c. 46      d. 49





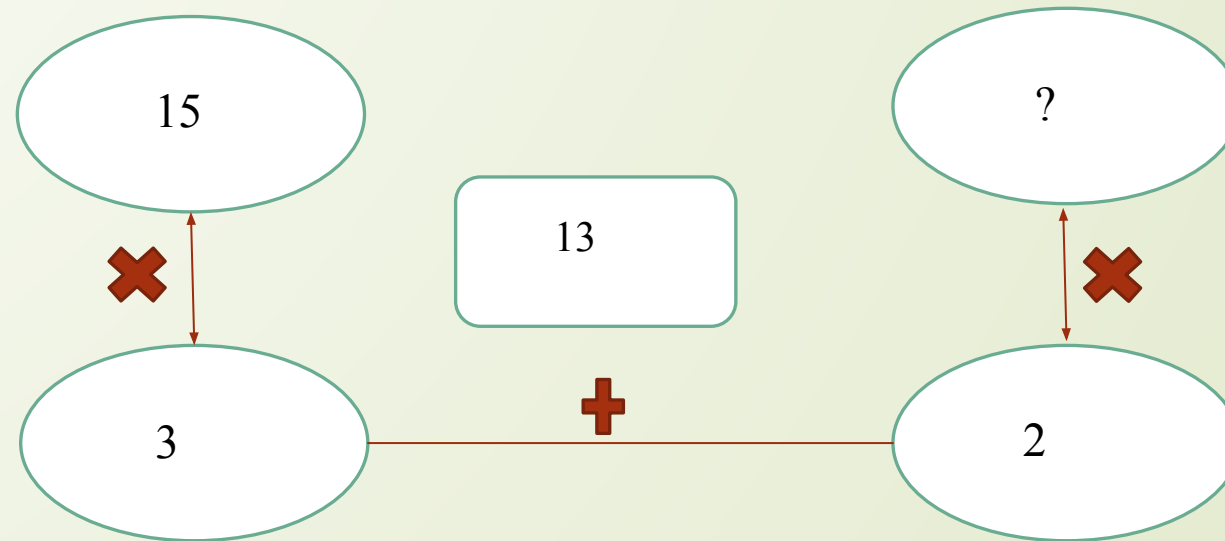
# Addition

**2. The average age of State level Cricket team of eleven is 22 years. The average age gets increased by 1 year when the coach is added, Find the Coach's age-**

**3. The average marks of a group of 10 people is 45 years. The average marks gets increased by 2 when a new person is added, Find the marks of the new person-**

# Weighted Average

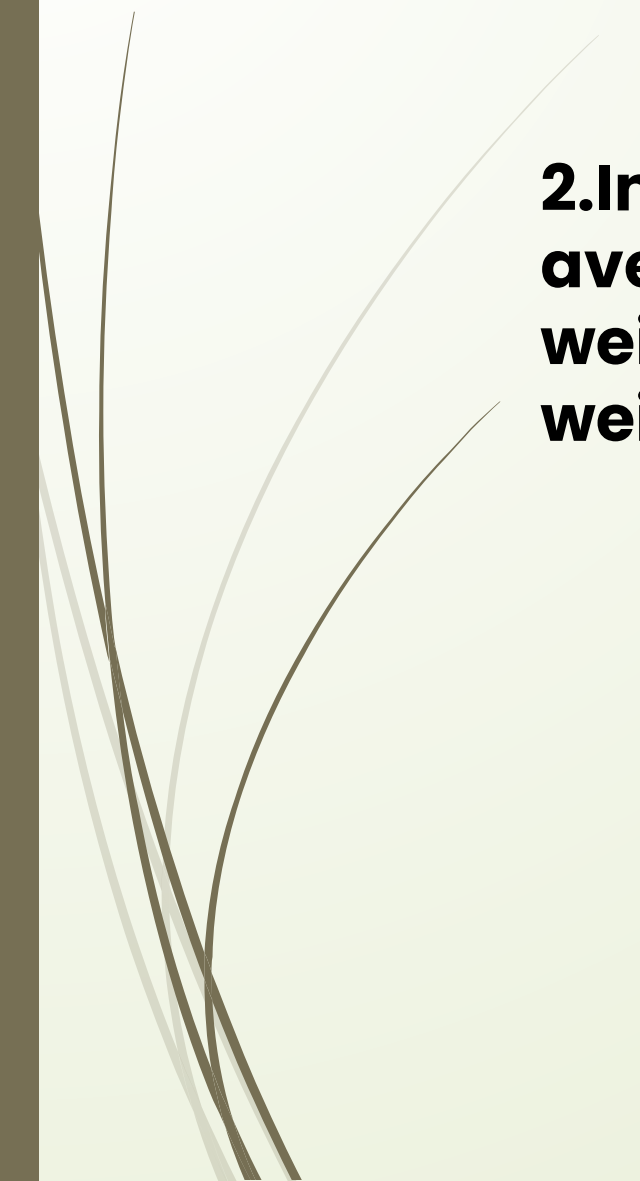
1. The average age of 30 girls is 13 yr. The average of first 18 girls is 15 yr. Find out the average age of remaining 12 girls.



$$\begin{aligned} (45 + 2x)/5 &= \\ 13 \\ x &= 10 \end{aligned}$$



# Weighted Average



**2. In a class of 39 students, there are 26 girls. The average weight of these girls is 42 kg and average weight of the full class is 48 kg. What is the average weight of the boys of the class?**